

Gut microbiota composition is associated with body weight, weight gain and biochemical parameters in pregnant women.



Obesity is associated with complications during pregnancy and increased health risks in the newborn. The objective of the present study was to establish possible relationships between gut microbiota, body weight, weight gain and biochemical parameters in pregnant women. Fifty pregnant women were classified according to their BMI in normal-weight (n 34) and overweight (n 16) groups. Gut microbiota composition was analysed by quantitative real-time PCR in faeces and biochemical parameters in plasma at 24 weeks of pregnancy. Reduced numbers of Bifidobacterium and Bacteroides and increased numbers of Staphylococcus, Enterobacteriaceae and Escherichia coli were detected in overweight compared with normal-weight pregnant women. E. coli numbers were higher in women with excessive weight gain than in women with normal weight gain during pregnancy, while Bifidobacterium and Akkermansia muciniphila showed an opposite trend. In the whole population, increased total bacteria and Staphylococcus numbers were related to increased plasma cholesterol levels. Increased Bacteroides numbers were related to increased HDL-cholesterol and folic acid levels, and reduced TAG levels. Increased Bifidobacterium numbers were related to increased folic acid levels. Increased Enterobacteriaceae and E. coli numbers were related to increased ferritin and reduced transferrin, while Bifidobacterium levels showed the opposite trend. Therefore, gut microbiota composition is related to body weight, weight gain and metabolic biomarkers during pregnancy, which might be of relevance to the management of the health of women and infants.